

Hybrid Modulation for an SHB-FB Converter Employed in an Integrated Onboard Charger and Auxiliary Power Module

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Abstract—This article presents a highly integrated electric vehicle (EV) charger that offers multiple charging modes, including a three-phase (3P) input 11-kW onboard charger (OBC), an 11-kW vehicle-to-load (V2L), and a 2.6-kW auxiliary power module (APM). The proposed multilevel, three-port topology simultaneously achieves magnetic and circuit integration while accommodating a 3P 400-V ac input and a 700-V dc bus using 650-V GaN HEMTs. A mathematical model of the zero voltage switching (ZVS) boundary of the stacked half-bridge–full-bridge (SHB-FB) converter in the proposed integrated charger is built. Based on this, one hybrid modulation (HM) method is further proposed to improve the full-range efficiency of the SHB-FB converter in the OBC and V2L modes. A prototype is built to validate the proposed integration method and the HM method. Experimental results validate the OBC, V2L, and APM modes of the proposed integrated charger and the effectiveness of the HM in improving the medium- and light-load efficiency by achieving full-range ZVS or quasi-ZVS (QZVS).

Index Terms—Auxiliary power module (APM), electric vehicle (EV), onboard charger (OBC), phase shift (PS) control, stacked half-bridge (SHB), zero-voltage switching (ZVS).

I. INTRODUCTION

DUE to their high performance, autonomy, zero tailpipe emissions, and low operating and maintenance cost, electric vehicles (EVs) have gradually replaced conventional vehicles powered by internal combustion engines (ICEs) and, therefore, have shown rapid development in the past decades [1]. One prevalent research topic is the search for a more convenient, efficient, and compact charging system. A typical EV charging system includes an onboard charger (OBC) and an auxiliary power module (APM), which is used to charge a high-voltage (HV) battery from the ac grid and a

low-voltage (LV) battery from the HV battery. Traditionally, these two units are designed, manufactured, and installed separately, resulting in large sizes and high costs, which, however, might be overcome by the integration of these two units [2], [3], [4], [5], [6], [7].

While mechanical integration is the most direct way by sharing the enclosure and cooling system between the OBC and the APM, circuit and magnetic integration can further reduce the cost and size. The block diagram of traditional standalone OBC and APM systems is shown in Fig. 1(a). The grid-side ac–dc converter is not shown in the figure as it is not the focus of this article. It is easy to see that the OBC and APM systems are connected to the HV battery and have a symmetric structure. Inspired by similarity, by sharing one half-bridge (HB), a partial circuit integration was achieved in [3]. In [8], an integrated battery charger was proposed by adding an extra circuit between the LDC and OBC side. However, such integration technology cannot save the usage of the magnetics and switches. As shown in Fig. 1(b), a more advanced magnetic and circuit integration of two systems is achieved by merging two traditional two-port transformers (XFMRs) into one three-port transformer that connects the active bridges on the primary dc bus, as well as HV and LV batteries [4], [5], [6], [9], [10]. In [4], [6], and [10], a triple active bridge (TAB) topology based on phase shift (PS) control is used, which adopted FB converters for all three ports. The typical schematic is shown in Fig. 2. Such a topology has the advantages of symmetrical and simplest structures, as the operation of the FB DAB converters has been thoroughly explored. However, the characteristics of the wide battery range, a large current at the LV side, and a large voltage ratio between HV and LV batteries are not optimized correspondingly. Zhu et al. [5] and Sha et al. [11] used a current-fed dual-active-bridge (CF-DAB) for LV battery sides, which reduced the transformer turn ratio from 33 to 12 and the transformer current rating was also reduced three times, which can simplify the transformer design. In [12], the zero-voltage switching (ZVS) boundary of the voltage-fed and current-fed DAB (VF-CF-DAB) was comprehensively analyzed. On the other hand, the emerging GaN switches are considered a good candidate for EV charger applications because of the advantage of lower R_{dson} , Q_g , E_{on} , and E_{off} . In [13], a 1-kW

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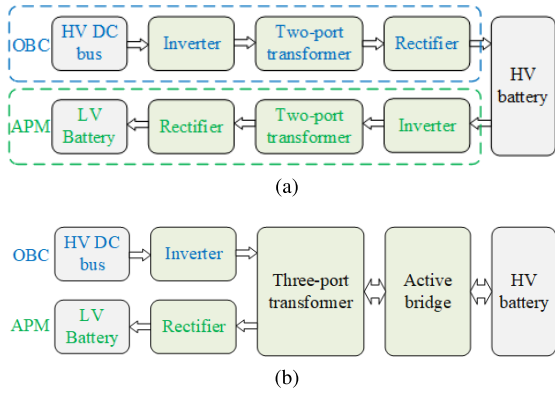


Fig. 1. Block diagram of different types of the EV charging system. (a) Traditional OBC and APM system. (b) Integrated OBC and APM system.

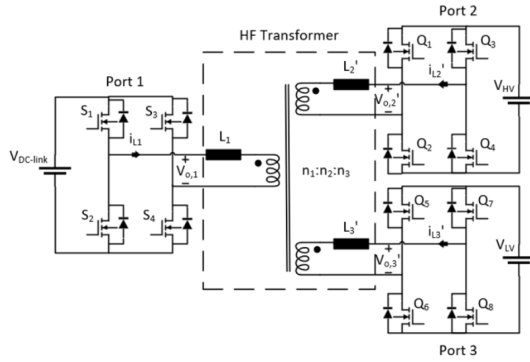


Fig. 2. Schematic of the typical TAB converter adopting FB on three ports.

GaN-based OBC using the traditional Totem-pole PFC and FB DAB is built. Compared to the Si-based prototype, the GaN-based charger shows better performance. In [14], a hybrid power switching circuit using Si MOSFET and GaN HEMT is proposed to achieve low switching and conduction loss. Based on the hybrid power switching circuit, an input-series and output-parallel (ISOP) modular-based OBC was built, where the good performance of GaN HEMTs in OBC applications was proved. However, the adoption of GaN HEMTs in the integrated OBC and APM system has not been explored a lot. Specifically, for the OBC with three-phase (3P) ac input, the dc link voltage can be over 800V. The traditional two-level (2L) topologies cannot accommodate the adoption of 650-V GaN HEMTs and the high-voltage dc link.

In terms of modulation methods of the HV PS converter, the traditional single PS (SPS) modulation method [15] suffers from losing ZVS at medium and light loads. Some other advanced modulation methods, for example, dual-PS (DPS) and triple-PS (TPS), have been proposed to achieve higher efficiency at medium and light load for DAB converters [16], [17], [18]. For EV charger applications, EV batteries usually have a relatively wide voltage range, for example, from 250 to 420 V for a 400-V battery pack, and the power needs to change between zero and full power to finish a complete charging cycle. However, most of them are only capable of realizing ZVS in a specific voltage or load range, and their analysis is based only on the traditional 2L full-bridge (FB) DAB converters. In [19], an SHB-SHB/FB converter with a resonant

tank and a reconfigurable SHB converter on the HV battery side was proposed to accommodate 800-V dc bus with 650-V GaN HEMTs. However, the proposed modulation method did not cover the full power range, and the wide-voltage-range ZVS is based on an LCL-T resonant network, which can increase the cost and complexity of the system and can be difficult to use in the integrated charger. From the literature review above, we can find that the full-range ZVS operation of the PS SHB-FB converter has not been thoroughly explored.

In this article, a new integrated EV charging system based on the three-part structure was proposed to achieve magnetic and circuit integration of the OBC and APM, as shown in Fig. 3. The traditional FB TAB topology was optimized to accommodate a 700-V dc bus effectively using 650-V GaN HEMTs, by changing the FB converter to the stacked half-bridge (SHB) converter, which uses fewer switches compared to other three-level (3L) structures and still has the same control freedom as the FB converter. The CF FB converter is used on the LV side to reduce the turn ratio between the HV and LV sides and the current stress on the LV side of the transformer. In addition, considering the wide voltage and wide power range of the OBC/V2L mode, the efficiency could be very low at the light load due to the hard switching of some switches in the SHB-FB stage. To improve the light-load efficiency, the ZVS condition of the SHB-FB converter is analyzed, and a hybrid modulation (HM) method is proposed to achieve full-range ZVS or quasi-ZVS (QZVS) for the SHB-FB converter in the OBC and vehicle-to-load (V2L) modes, which leads to high efficiency in the full range compared to the traditional modulation method. Such improvement is also important to ensure the stable operation of such a GaN-based charging system by avoiding the thermal runaway problem, which is particularly serious due to the small package of GaN HEMTs. This article is organized as follows. Section II introduces the topology and basic operation mode of the proposed integrated charger. Section III presents the ZVS analysis and the proposed HM. Section IV is the experimental results to validate both the proposed integrated charger and the HM. Section V concludes.

II. PROPOSED TOPOLOGY AND THE OPERATION MODE

The proposed system includes one 3P ac–dc converter that bridges the grid and the dc bus and one TAB converter that bridges the dc bus, an HV battery, and an LV battery. To build a system with 700 V-dc bus using only 650-V GaN HEMTs, two different three-level structures, including the active neutral-point-clamping (ANPC) [20] and SHB switching circuits, are used for ac–dc and dc–dc stages, respectively.

A. Topology Introduction

As shown in Fig. 3, the front-end ac–dc stage consists of three 3L ANPC phase legs (Q_{x1} – Q_{x6}) and works as a power factor conversion (PFC) and an inverter, respectively, in the OBC mode and the V2L mode. An SHB converter (P_1 – P_{x4}) is connected after the ac–dc stage to bridge the dc bus (C_{dc1} and C_{dc2}) and the primary port of the transformer, and the blocking capacitor (C_b) is used to block the dc component

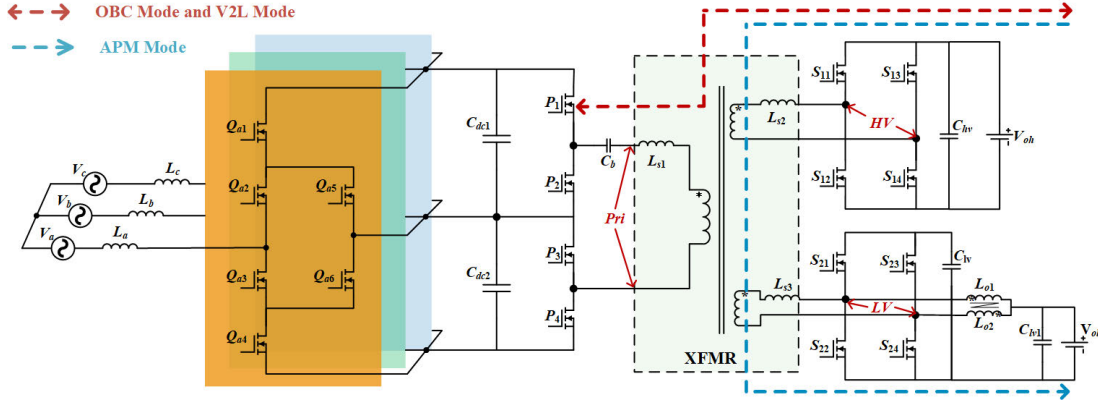


Fig. 3. Proposed integrated onboard charging system.

from the generated voltage of the SHB converter and ensure that only the ac component is applied to the primary terminal. Three ports of the transformer are named *Pri*, HV, and LV, respectively, and L_{s1} – L_{s3} are the leakage inductance on them, respectively. A VF–FB converter (S_{11} – S_{14}) is connected to the HV port of the transformer. A CF–FB converter (S_{21} – S_{24}) with a coupled inductor (L_{o1} – L_{o2}) are adopted for the LV port [21].

Compared to other 3L topologies, such as 3L NPC and ANPC converters, fewer devices are needed in the SHB converter, as shown in Fig. 4, where v_{SHB_out} and v_{Pri} represent the voltage generated by the SHB converter and the ac voltage on the primary port of the transformer, respectively, and v_{Pri} represents the ac voltage on the HV port of the transformer. The magnitude of v_{Pri} and v_{HV} are V_P and V_H , respectively. L_s is the equivalent leakage inductance reflected to the primary port. The current that flows into the primary winding defines the positive polarity of the leakage inductor current, i_{L_s} , on both ports. v_{SHB_out} can be calculated as follows:

$$v_{SHB_out} = v_{ad} - v_{bd} \quad (1)$$

where

$$v_{ad} = v_{ac} + v_{cd} = v_{ac} + \frac{V_{dc_link}}{2} \quad (2)$$

by combining (1) and (2), we can get

$$v_{SHB_out} = \frac{V_{dc_link}}{2} + v_{ac} - v_{bd} \quad (3)$$

from (3), with the dc component blocked by C_b , the voltage across the primary port can be expressed as

$$v_{Pri} = v_{ac} - v_{bd} \quad (4)$$

from (4), when inserting the PS between two HB legs of the SHB converter, v_{Pri} can be 3L ac voltage, which means that the SHB converter with blocking capacitors has the same control freedom as the FB converter.

B. Operation Modes and Basic Principles

When working in the OBC and V2L modes, power transfer is between the 3P ac and the HV battery. The front-end ac–dc converter bridges the 3P ac main and dc bus, incorporating active voltage balancing control and maintaining a fixed voltage 700 V. The SHB–FB converter regulates the charging

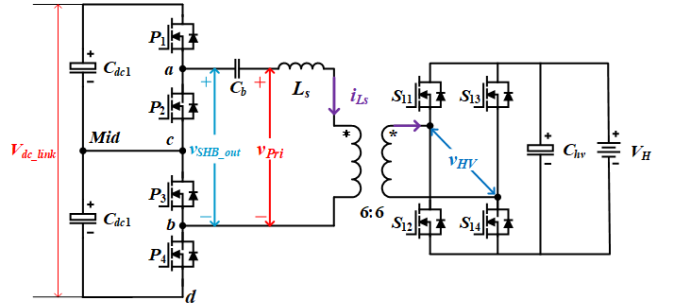


Fig. 4. Schematic of the SHB converter.

current and power. When working in the APM mode, power transfer is from the HV battery to the LV battery.

The main control variables and typical waveforms of the OBC and APM mode are shown in Fig. 5.

In the OBC and V2L modes, as shown in Fig. 5(a), the parameter α and β are used to regulate the inner PSs of the *Pri* and HV sides, respectively, and ϕ is the PS between the *Pri* and HV sides. All switches have a fixed 50% duty cycle and any two switches in an HB leg are always complimentary. The parameters α and β are defined as the time differences between P_1 and P_3 and between P_1 and S_{13} , respectively, divided by $T_s/2$. Similarly, ϕ is defined as the time difference between P_1 and S_{11} , also divided by $T_s/2$. Based on the definition above, when α and β are equal to 1 and $1 + \phi$, respectively, there is no inner PS on both sides, which is just the SPS modulation. When they are different from 1 and $1 + \phi$, respectively, the DPS or TPS modulation is applied on the SHB–FB converter.

The main control variables and typical waveforms of the APM mode are shown in Fig. 5(b). In the APM mode, combined duty cycle and PS modulation methods proposed in [21] are used to achieve ZVS in the wide voltage range and reduce the switching current. On both sides, the PS between two half-bridges (HBs) is always 1, where the PS is defined as the time difference between the top switches of two HBs divided by the half period. Based on Fig. 5(b), it means that

$$t_4 - t_0 = t_5 - t_1 = T_s/2. \quad (5)$$

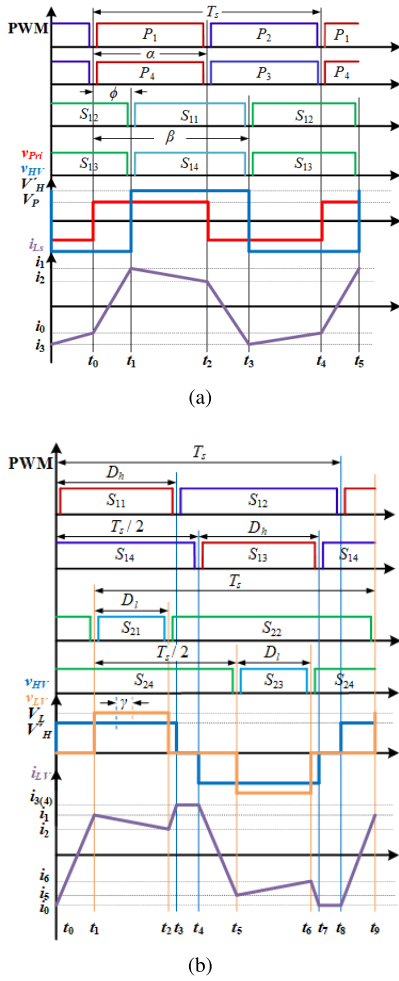


Fig. 5. Main control variables and typical transformer voltage and current waveform of different operation modes. (a) SHB-FB converter in OBC and V2L modes. (b) VF-CFDB converter in the APM mode.

The PS, γ , between two sides is defined as the time difference between the center of two ac voltages, v_{HV} and v_{LV} in Fig. 5, divided by half period, which can be expressed as

$$\gamma = (t_1 + t_2 - t_3)/T_s. \quad (6)$$

The duty cycles, D_h and D_l in Fig. 5, represent the PWM width of the upper switches, which are defined as

$$D_h = \frac{t_3 - t_0}{T_s/2} \quad (7)$$

$$D_l = \frac{t_2 - t_1}{T_s/2}. \quad (8)$$

The duty cycles of the upper switches of two HBs on the same side, which are S_{11} and S_{13} ; S_{21} and S_{23} are always the same. Since the APM modulation method is not the main contribution of this article, it will not be introduced in detail in this article, and only the key characteristics are summarized as follows.

- 1) The operation range can be divided into two modes based on the relationship of the PS and the duty cycles, which can be expressed as

$$\begin{aligned} \text{Mode 1: } \gamma &\leq D_h - D_l \\ \text{Mode 2: } \gamma &> D_h - D_l. \end{aligned} \quad (9)$$

- 2) In Mode 1, the output power is only related to the PS, and the voltages but not related to the duty cycle.
- 3) In Mode 1, the switching currents of both HV and LV ports are only related to their own duty cycle, respectively, but not related to the PS, which means that the switching current of each port can be regulated and optimized by the corresponding duty cycle independently.
- 4) Mode 1 is preferred due to the good features b) and c). In mode 2, the output power and the switching currents are related to multiple variables. Such coupled relationships can lead to challenges in realizing the ZVS and optimizing the switching current.

III. ZVS ANALYSIS FOR THE SHB-FB CONVERTER

A. ZVS Condition

To turn on switches with ZVS, the turning-on current needs to have negative polarity, which means that the current flows from the source to the drain and has a large enough magnitude to charge and discharge the output capacitors of the corresponding switches completely. In the SHB-FB converter, ZVS is realized by the leakage inductance current. The minimum magnitude of the current to achieve ZVS can be expressed as follows [22]:

$$I_r \geq \sqrt{\frac{2C_{\text{oss,eq}} V_{ds}^2}{L_s}} \quad (10)$$

where V_{ds} is the drain-source voltage, $C_{\text{oss,eq}}$ is the equivalent output capacitance, and L_s is the leakage inductance reflected to the corresponding side.

The polarity of the inductor current to achieve the ZVS for all switches in the SHB-FB converter can be analyzed by the case of applying SPS control at heavy load as shown in Fig. 5. The transition process in Fig. 5 from t_0 to t_3 is depicted in Fig. 6, where t_n^- , t_n , and t_n^+ represent before, during, and after the dead-band time, respectively. It should be noted that the MOSFET symbol is used to represent GaN HEMTs in Figs. 3 and 5 to show the reverse conducting easily. The complete gray and black switches represent the condition of off and on, respectively, and the gray switch with black anti-parallel diode represents the reverse conducting. The details of the transition process are summarized as follows.

1) P_1 and P_4 : At t_0^- , the inductor current is negative, P_2 , P_3 , S_{12} , and S_{13} are on and P_2 and P_3 are going to be turned off. At t_0 , P_2 and P_3 are off and P_1 and P_4 are reverse conducting. At t_0^+ , P_1 and P_4 turn on with ZVS achieved, and S_{12} and S_{13} keep on.

2) S_{11} and S_{14} : From t_0^+ to t_1^- , the polarity of the inductance current changes from negative to positive. At t_1^- , P_1 , P_4 , S_{12} , and S_{13} are on and S_{12} and S_{13} are going to be turned off. At t_1 , S_{12} and S_{13} turn off and S_{11} and S_{14} are reverse conducting. At t_1^+ , S_{11} and S_{14} turn on with ZVS achieved and P_1 and P_4 keep on.

3) P_2 and P_3 : At t_2^- , the inductor current is still positive, P_1 , P_4 , S_{11} , and S_{14} are on and P_1 and P_4 are going to be turned off. At t_2 , P_1 and P_4 are off and P_2 and P_3 are reverse conducting. At t_2^+ , P_2 and P_3 turn on with ZVS achieved, and S_{11} and S_{14} keep on.

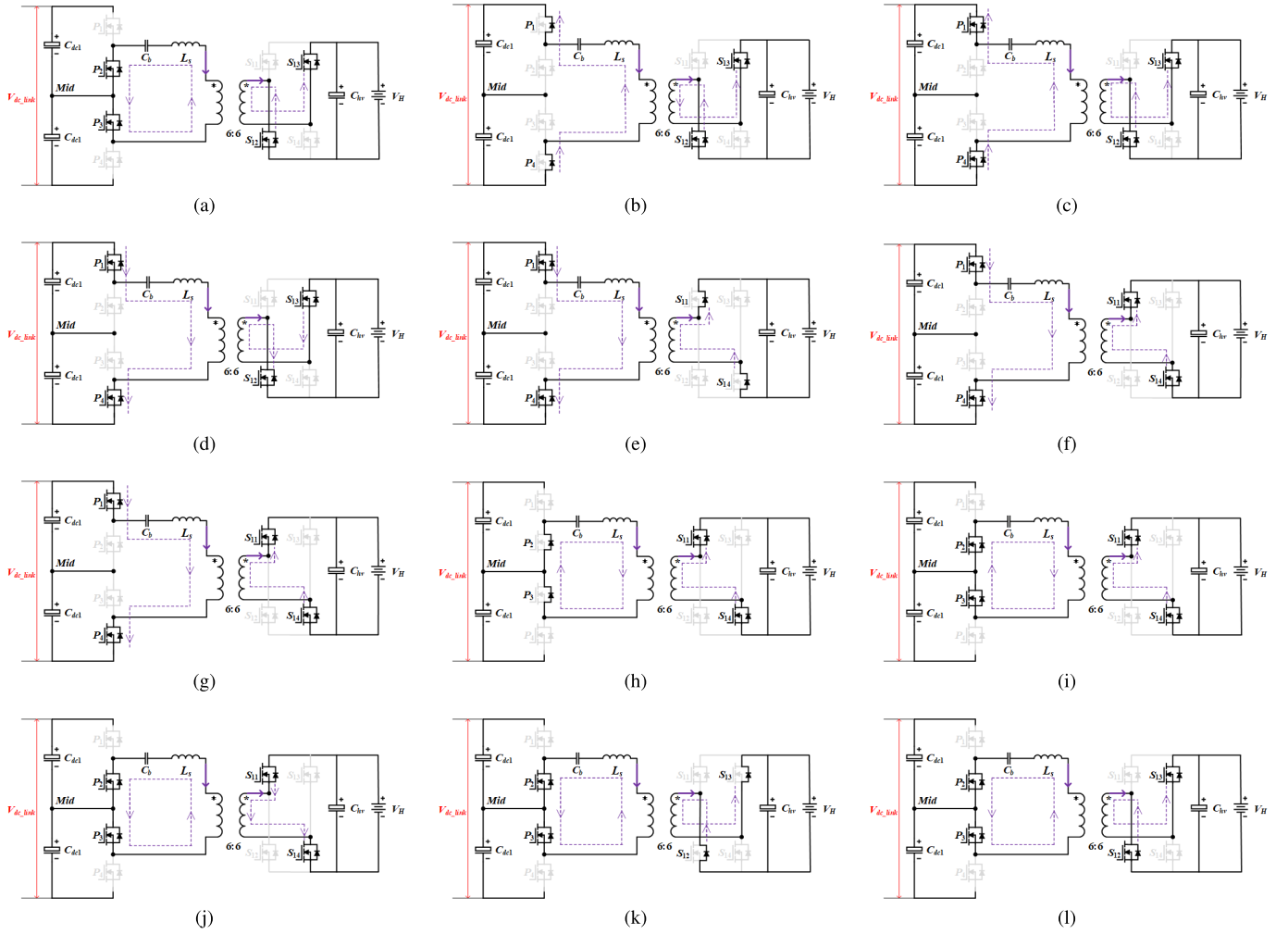


Fig. 6. Transition process of the half-cycle on the primary side. (a) t_0^- . (b) t_0 . (c) t_0^+ . (d) t_1^- . (e) t_1 . (f) t_1^+ . (g) t_2^- . (h) t_2 . (i) t_2^+ . (j) t_3^- . (k) t_3 . (l) t_3^+ .

4) S_{12} and S_{13} : From t_2^+ to t_3^- , the polarity of the inductance current changes from positive to negative. At t_3^- , P_2 , P_3 , S_{11} , and S_{14} are on and S_{11} and S_{14} are going to be turned off. At t_3 , S_{11} and S_{14} are off and S_{12} and S_{13} are reverse conducting. At t_3^+ , S_{12} and S_{13} turn on with ZVS achieved, and P_2 and P_3 keep on.

From the analysis above, the required current polarity of the SHB-FB converter can be summarized as that the inductor current needs to be negative for P_1 , P_4 , S_{12} , and S_{13} and positive for P_2 , P_3 , S_{11} , and S_{14} , respectively, and the magnitude needs to follow the requirement represented by (10).

It is clear that the requirements of ZVS achievement can be satisfied by the SPS modulation at heavy load; however, when the power reduces, the current polarity of some switches will change using SPS modulation, which will lead to the hard switching-on. The OBC for EVs requires the capability to run in the full-battery-voltage range and the full-power range. Hard switching on at some operation points can harm the overall efficiency and lead to the risk of damage to switches. To keep ZVS, DPS or TPS needs to be applied to ensure the right polarity for ZVS achievement. The boundary of different modulations and the needed PS are analyzed in the following, which is divided into two parts on the basis of the relationship

of voltages on the primary side, V_P , and the reflected voltage on the primary side from the HV side, V'_H .

B. When $V_P > V'_H$

1) State 4: When $V_P > V'_H$, the switching current using the SPS control is shown in Fig. 7(a) and can be expressed as follows:

$$\begin{cases} i_0 = -\frac{T_s}{4L_s}[V_P + (2\phi - 1)V'_H] \\ i_1 = \frac{T_s}{4L_s}[(2\phi - 1)V_P + V'_H] \end{cases} \quad (11)$$

where T_s is the switching period and L_s is the leakage inductance. With power decreases, the magnitude of the switching current decreases. Therefore, the magnitude of the switching current of the primary switches i_1 and i_3 will first reach the minimum ZVS current I_r . Based on the requirement of $i_1 \geq I_r$, the boundary condition for full ZVS operation under the SPS control can be generated as

$$\begin{cases} \alpha = 1 \\ \beta = 1 + \phi \\ \frac{4L_s f_s I_{r1} + V_P - V'_H}{2V_P} \leq \phi \leq 0.5. \end{cases} \quad (12)$$

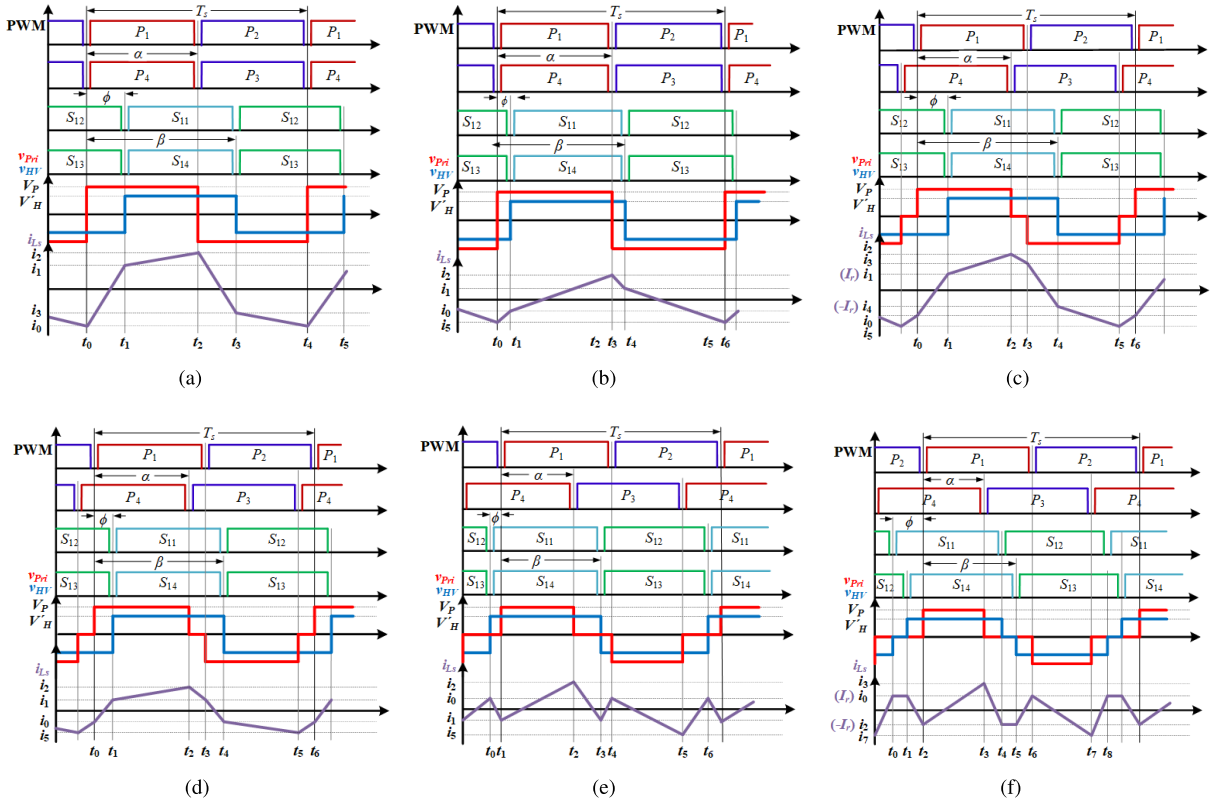


Fig. 7. PWM and typical waveform of the SHB-FB converter at different states when $V_P > V'_H$. (a) State 4: SPS, ZVS. (b) State 4.1: SPS, HS. (c) State 3: PDPS, ZVS. (d) State 2: PDPS, $\phi > 0$, QZVS. (e) State 1: PDPS, $\phi < 0$, QZVS. (f) State 0: TPS, ZVS.

2) *State 4.1*: Based on (11), with ϕ decreasing, if still using the SPS modulation, the magnitude of i_1 and i_3 will be smaller than I_r and even change the polarity. Therefore, all secondary switches, S_{11} – S_{14} , will lose ZVS, as shown in Fig. 7(b), which will lead to lower efficiency. To avoid this, the primary DPS (PDPS) takes over after ϕ reaches the boundary of State 4.0.

3) *State 3*: With decreasing power, when the magnitude of i_1 and i_3 is equal to I_r , PDPS takes over, α starts to reduce, as shown in Fig. 7(c). At this stage, the switching current can be expressed as

$$\begin{cases} i_0 = -\frac{T_s}{4L_s}[\alpha V_P + (2\phi - 1)V'_H] \\ i_1 = \frac{T_s}{4L_s}[(2\phi - \alpha)V_P + V'_H] \\ i_2 = \frac{T_s}{4L_s}[\alpha V_P + (1 - 2\alpha + 2\phi)V'_H] \end{cases} \quad (13)$$

where α is regulated to ensure the ZVS of secondary switches by fixing i_1 to I_r . The ZVS boundary and needed PS can be calculated as

$$\begin{cases} \alpha = \frac{V'_H - 4L_s f_s I_{r1}}{V_P} + 2\phi \\ \beta = 1 + \phi \\ \frac{4L_s f_s I_{r1}}{V_P + V'_H} \leq \phi < \frac{4L_s f_s I_{r1} + V_P - V'_H}{2V_P} \end{cases} \quad (14)$$

4) *State 2*: When $i_0 = -I_r$, the system starts another stage, as shown in Fig. 7(d), where only P_3 and P_4 can achieve ZVS, all others can achieve QZVS, that is, QZVS, which happens

when the switching current has right polarity but is not large enough to fully discharge the drain–source capacitance of the corresponding switches. At this stage, the switching current can still be expressed as (13). However, α is regulated to keep i_0 equal to i_1 , and will decrease together with power decreases until they are equal to zero when ϕ reduces to zero. Therefore, the ZVS boundary and the needed PS at this stage can be calculated as follows:

$$\begin{cases} \alpha = \phi + \frac{(1 - \phi)V'_H}{V_P} \\ \beta = 1 + \phi \\ 0 \leq \phi < \frac{4L_s f_s I_{r1}}{V_P + V'_H} \end{cases} \quad (15)$$

5) *State 1*: After ϕ reaches 0 and continues to decrease, ϕ , i_0 , and i_1 both change polarity and increase in reverse, as shown in Fig. 7(e). It is worth mentioning that the power is still flowing from the primary side to the HV side. The switching currents at this stage can be expressed as follows:

$$\begin{cases} i_0 = \frac{T_s}{4L_s}(V'_H - \alpha V_P) \\ i_1 = \frac{T_s}{4L_s}[-\alpha V_P + (1 + 2\phi)V'_H] \\ i_2 = \frac{T_s}{4L_s}[\alpha V_P + (1 + 2\phi - 2\alpha)V'_H] \end{cases} \quad (16)$$

At this stage, α is still regulated to keep i_0 equal to i_1 , and their magnitude will increase together with ϕ decreasing until they are both equal to I_r . Therefore, the ZVS boundary and

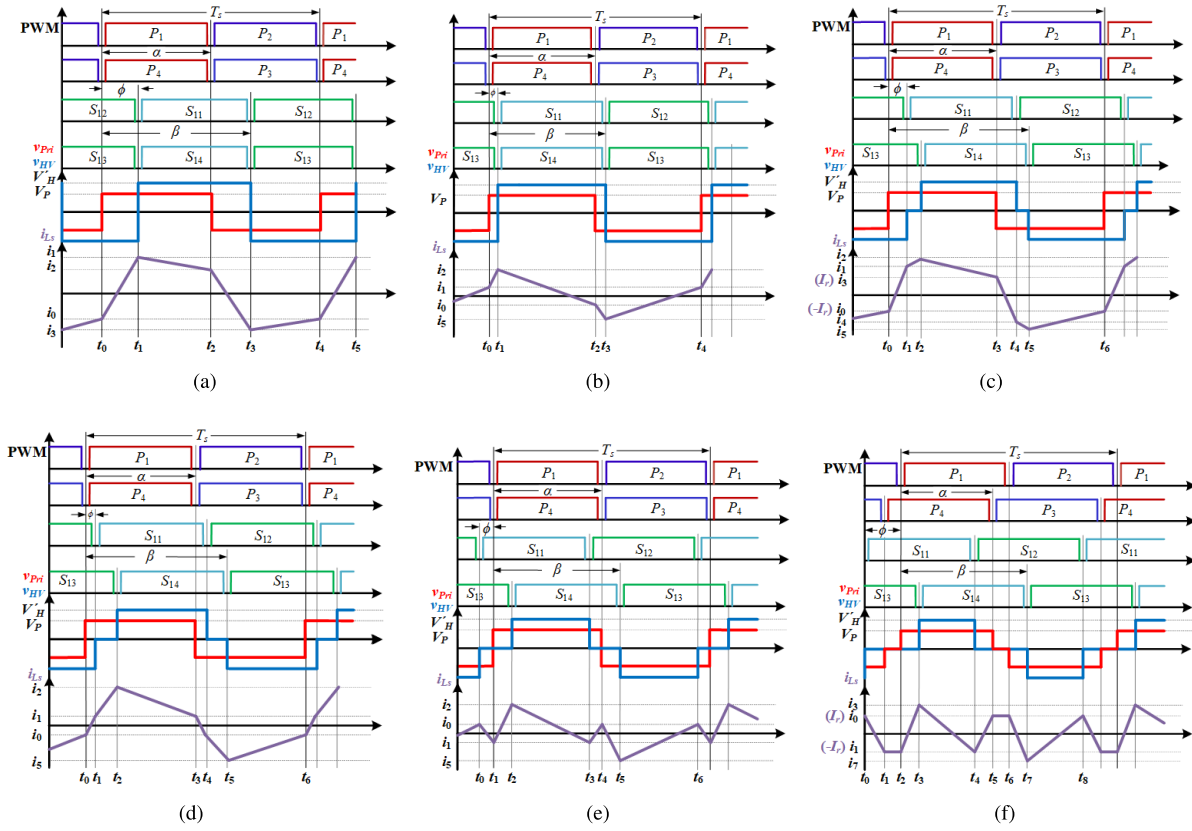


Fig. 8. PWM and typical waveform of the SHB-FB converter at different states when $V_P < V'_H$. (a) State 4: SPS, ZVS. (b) State 4.2: SPS, HS. (c) State 5: SDPS, ZVS. (d) State 6: SDPS, $\phi > 0$, QZVS. (e) State 7: SDPS, $\phi < 0$, QZVS. (f) State 8: TPS, ZVS.

needed PS can be calculated as

$$\begin{cases} \beta = 1 + \phi \\ \alpha = \frac{(1 + \phi)V_H}{V_P} \\ -\frac{4L_s f_s I_{r1}}{V'_H} \leq \phi < 0. \end{cases} \quad (17)$$

6) *State 0*: When the magnitude of i_0 and i_1 reach I_r again, all switches can achieve ZVS again. When power decreases, the TPS will take over, as shown in Fig. 7(f). The switching current using TPS modulation can be expressed as follows:

$$\begin{cases} i_0 = \frac{T_s}{4L_s} [(-\alpha)V_P + (2 + \phi - \beta)V'_H] \\ i_1 = i_0 \\ i_2 = \frac{T_s}{4L_s} [-\alpha V_P + (\phi + \beta)V'_H] \\ i_3 = \frac{T_s}{4L_s} [\alpha V_P + (\phi + \beta - 2\alpha)V'_H] \\ i_4 = \frac{T_s}{4L_s} [\alpha V_P + (\beta - \phi - 2)V'_H]. \end{cases} \quad (18)$$

At this stage, α and β are regulated to fix the magnitudes of i_0 and i_1 to I_r , which can reduce the hard turn-off loss by avoiding switching current to keep increasing when ϕ decreases. At the same time, the magnitude of i_2 decreases with ϕ decreasing until the power is equal to zero. Based on the control goal, the needed inner PS can be generated by

solving the equations as follows:

$$\begin{cases} \Delta i_{12} = -2I_r \\ \Delta i_{23} = -\Delta i_{34}. \end{cases} \quad (19)$$

The ZVS boundary and needed PS are calculated as

$$\begin{cases} \alpha = \frac{V'_H(1 + \phi)}{V_P} \\ \beta = 1 - \frac{4L_s f_s I_r}{V'_H} \\ -0.5 \leq \phi \leq -\frac{4L_s f_s I_{r1}}{V'_H}. \end{cases} \quad (20)$$

C. When $V_P < V'_H$

1) *State 4*: When $V_P < V'_H$, the inductor current is depicted in Fig. 8(a), and the expression of the switching current is the same as the case where $V_P > V'_H$, which is shown in (11). With power decreases, the magnitude of i_0 and i_2 will reach I_r , earlier. The ZVS condition with the SPS control can be generated as

$$\begin{cases} \alpha = 1 \\ \beta = 1 + \phi \\ \frac{4L_s f_s I_{r1} - V_P + V'_H}{2V'_H} \leq \phi \leq 0.5. \end{cases} \quad (21)$$

2) *State 4.2*: Based on (11), with ϕ decreasing, if still using the SPS modulation, the magnitude of i_0 and i_2 will be smaller than I_r and even change the polarity. Therefore, all secondary switches, P_1 – P_4 , will lose ZVS, as shown in Fig. 8(b), which will lead to lower efficiency. To avoid this, the secondary DPS (SDPS) takes over after ϕ reaches the boundary of State 4.0.

3) *State 5*: When power decreases and the magnitude of i_0 and i_2 reaches I_r , SDPS control takes over. The typical waveform is shown in Fig. 8(c). The switching current can be expressed as

$$\begin{cases} i_0 = -\frac{T_s}{4L_s} [V_P + (\phi + \beta - 2)V'_H] \\ i_1 = \frac{T_s}{4L_s} [(2\phi - 1)V_P + (\phi - \beta + 2)V'_H] \\ i_2 = \frac{T_s}{4L_s} [(2\beta - 3)V_P + (\phi - \beta + 2)V'_H] \end{cases} \quad (22)$$

where β is regulated to fix the magnitude of the switching current of switches on the primary side, i_0 and i_3 , to I_r . Therefore, the ZVS condition of stage 5 can be calculated as follows:

$$\begin{cases} \alpha = 1 \\ \beta = \frac{4L_s f_s I_{r1} - V_P}{V'_H} + 2 - \phi \\ \frac{4L_s f_s I_{r1}}{V_P + V'_H} \leq \phi < \frac{4L_s f_s I_{r1} - V_P + V'_H}{2V'_H}. \end{cases} \quad (23)$$

4) *State 6*: When i_1 and i_4 are equal to I_r , with power decreases, only S_{11} and S_{14} can achieve ZVS, while all others can achieve QZVS. During this stage, β is regulated to keep the magnitudes of i_0 , i_1 , i_3 , and i_4 the same and make them decrease together with the decrease of power. The current expression is still (22), as shown in Fig. 8(d). Based on the control goal, the ZVS condition, and needed PSs of stage 6 can be calculated as

$$\begin{cases} \alpha = 1 \\ \beta = 2 - \frac{(1 - \phi)V_P}{V'_H} \\ 0 \leq \phi < \frac{4L_s f_s I_{r1}}{V_P + V'_H}. \end{cases} \quad (24)$$

5) *State 7*: Based on (22), when $\phi = 0$, both i_0 and i_1 both are equal to 0. Then, with power continued to decrease, ϕ , i_0 , and i_1 will change polarity and increase reversely, as shown in Fig. 8(e). The switching currents at this stage can be expressed as follows:

$$\begin{cases} i_0 = \frac{T_s}{4L_s} [-(2\phi + 1)V_P + (2 + \phi - \beta)V'_H] \\ i_1 = \frac{T_s}{4L_s} [-V_P + (2 + \phi - \beta)V'_H] \\ i_2 = \frac{T_s}{4L_s} [(2\beta - 3)V_P + (2 + \phi - \beta)V'_H] \end{cases} \quad (25)$$

where β is still used to achieve $i_0 + i_1 = 0$, however, with ϕ decreasing, the magnitude of these currents will increase until they are both equal to I_r . Therefore, the ZVS condition of this

stage can be calculated as

$$\begin{cases} \alpha = 1 \\ \beta = \frac{-(1 + \phi)V_P + (2 + \phi)V'_H}{V'_H} \\ -\frac{4L_s f_s I_r}{V_P} \leq \phi < 0. \end{cases} \quad (26)$$

6) *State 8*: When the magnitudes of i_0 and i_1 are equal to I_r , they will be fixed by using the TPS modulation, as shown in Fig. 8(f). The ZVS is achieved for all switches until the load decreases to zero. The inductance current using TPS modulation can be expressed as follows:

$$\begin{cases} i_0 = \frac{T_s}{4L_s} [(\alpha - 2\phi - 2)V_P + (2 + \phi - \beta)V'_H] \\ i_1 = \frac{T_s}{4L_s} [(-\alpha)V_P + (2 + \phi - \beta)V'_H] \\ i_2 = i_1 \\ i_3 = \frac{T_s}{4L_s} [(-\alpha + 2\beta - 2)V_P + (2 + \phi - \beta)V'_H]. \end{cases} \quad (27)$$

Based on the control goal, the ZVS condition and needed PS are generated as

$$\begin{cases} \alpha = 1 + \phi + \frac{4L_s I_r f_s}{V_P} \\ \beta = 2 + \phi - \frac{V_P(\phi + 1)}{V'_H} \\ -0.5 \leq \phi < -\frac{4L_s f_s I_r}{V_P}. \end{cases} \quad (28)$$

D. Boundary Condition and State Transition

The proposed HM method divides the entire power range into 9 different states by ϕ . With the power decreasing from the peak value to zero, the PS ϕ also decreases. The boundary conditions between different states are summarized in Table I, where Cd4/3 means the condition from State 4 to State 3. The expression of the boundary condition from State 3 to State 4 is the same; it just needs to change the less than sign to the greater than or equal sign. The smooth change among these states is important. The power expression in different states is summarized in Table II, based on which the relationship between three PSs and power is shown in Fig. 9. All PSs are found to change smoothly with power. In closed-loop control, ϕ is generated by the PI controller and the state will be determined by the boundary of ϕ . Meanwhile, the other two PSs can be calculated by the mathematical model.

E. Implementation of HM

The block diagram of the implementation layout of the proposed HM is shown in Fig. 10. In the EV charging system, the OBC needs to receive the charging command, I_{cmd} , from the MCU as the reference input of the PI controller of the OBC. With the output current sensing result, I_o , as the feedback of the PI controller, the PI controller will generate the needed PS, ϕ . Coordinating with the measured dc link voltage, V_{dc_link} , and the HV battery voltage, V_{HVB} , the state will be determined by the ZVS boundary model developed in

TABLE I
BOUNDARY CONDITIONS BETWEEN ADJACENT STATES

Cd	Expression
Cd4/3	$\phi < \frac{4L_s f_s I_{r1} + V_P - V_{H'}}{2V_P}$
Cd3/2	$\phi < \frac{4L_s f_s I_{r1}}{V_P + V_{H'}}$
Cd2/1	$\phi < 0$
Cd1/0	$\phi < -\frac{4L_s f_s I_{r1}}{V_{H'}}$
Cd4/5	$\phi < \frac{4L_s f_s I_{r1} - V_P + V_{H'}}{2V_{H'}}$
Cd5/6	$\phi < \frac{4L_s f_s I_{r1}}{V_P + V_{H'}}$
Cd6/7	$\phi < 0$
Cd7/8	$\phi < -\frac{4L_s f_s I_r}{V_P}$

TABLE II
RELATIONSHIP OF POWER AND PSS

State	Power expression
0	$\frac{V_P V_{H'}}{4L_s f_s} (\phi + \beta - \alpha) \alpha$
1	$\frac{V_{H'}^2 (1 + \phi)}{4L_s f_s V_P} [(1 + \phi)(V_P - V_{H'})]$
2 & 3	$\frac{V_P V_{H'}}{4L_s f_s} [-2\phi^2 - \alpha^2 + 2\alpha\phi + \alpha]$
4	$\frac{V_P V_{H'} \phi (1 - \phi)}{2L_s f_s}$
5 & 6	$\frac{V_P V_{H'}}{4L_s f_s} (-\phi^2 - \beta^2 + 3\beta + \alpha - 2)$
7	$\frac{V_P}{4L_s f_s} [(\phi^2 + \phi)V_P + (2 + \phi - \beta)(\beta - 1)V_{H'}]$
8	$\frac{V_P}{4L_s f_s} [(\phi^2 - \phi\alpha + 2\phi - \alpha + 1)V_P + (2 + \phi - \beta)(\beta - 1)V_{H'}]$

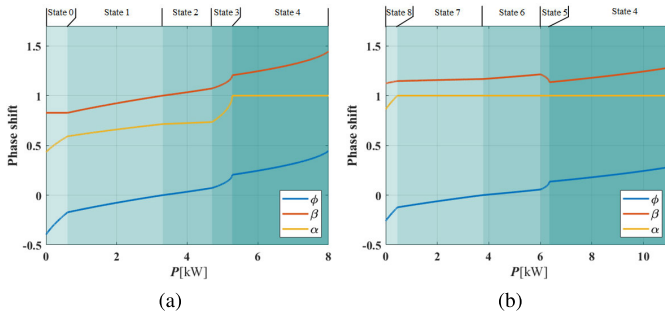


Fig. 9. Output power versus PS with different output voltages. (a) $V_{H'} = 250$ V. (b) $V_{H'} = 420$ V.

Sections III-B and III-C. The PS calculator can then generate the other two PSs α and β . Based on the generated three PSs, PWM signals are generated by the PWM generator. It is worth mentioning that the state can only transition to the adjacent states. For example, when $V_P > V_{H'}$, the system will start from state 0, go through states 1, 2, and 3, until state 4 with the

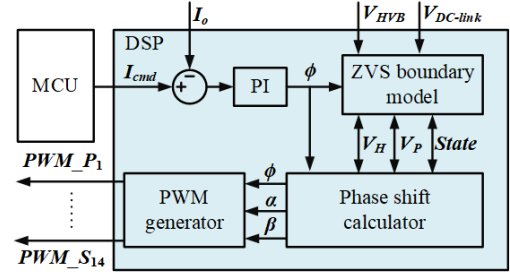


Fig. 10. Implementation layout of the proposed HM.

TABLE III
SYSTEM SPECIFICATIONS

Parameters/ Specifications	Value/ Part Number
AC input line-line voltage ($V_{AC_LL_rms}$)	400 V
DC link voltage (V_{DC_link})	700 V
HV Battery voltage (V_{HVB})	250 V - 420 V
LV battery voltage (V_{LVB})	9-16 V
Switching frequency of the OBC mode (f_{s_OBC})	150 kHz
Switching frequency of the APM mode (f_{s_APM})	100 kHz
Transformer turn ratio	6:6:1
HV switches	GS65516T
LV switches	GS61008T
Number of paralleled HV switches	3
Number of paralleled LV switches	4
Volumn of the prototype	$457 \times 295 \times 70 \text{ mm}^3 = 9.44L$
Power density	2.61 kW/L

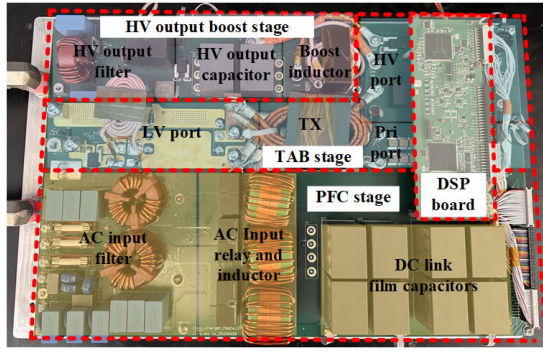
increase of the output current if the needed ϕ is larger than the ZVS boundary based on the PI controller. The smooth transition of all PSs with the change of power, as shown in Fig. 9, can ensure that no big step change happens when the state changes.

IV. EXPERIMENTAL VERIFICATION

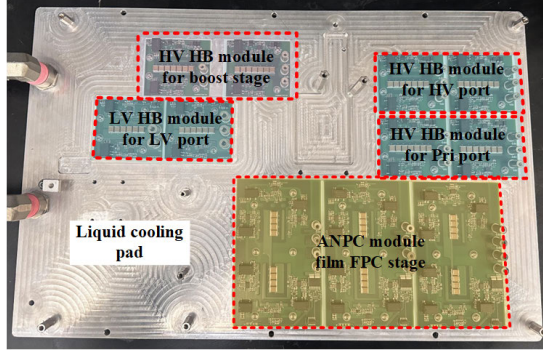
To validate the proposed integrated charger solution and the HM methods of the SHB-FB DCDC converter, three different power modules were designed, including a 3L ANPC module, a 2L HV HB module, and a 2L LV HB module, which integrate GaN devices, a gate driver circuit, and decoupling capacitors. Using these modules, a multifunctional integrated charger was built, including the 11 kW OBC mode, the 11 kW V2L mode, and the 2.6-kW APM mode, as shown in Fig. 11. The system specifications are shown in Table III. The prototype includes three layers, that is, the motherboard and control card on top, power modules in the middle, and a liquid cooling plate on the bottom. The volume of the prototype, including the heat sink, is 9.44 L.

A. OBC and V2L Mode

The typical waveform of the ac/dc and dc/dc stages in the 11-kW OBC and V2L mode is shown in Fig. 12. A battery



(a)



(b)

Fig. 11. GaN-based integrated charger prototype. (a) DSP board and Mother-board with magnetic components and capacitors. (b) Switching modules and the liquid cooling plate.

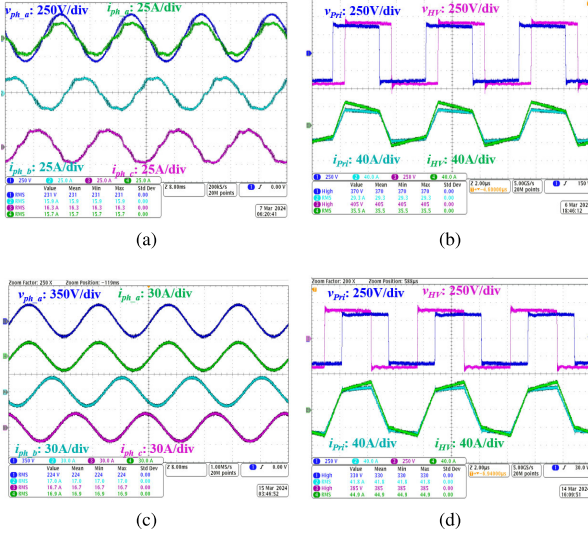


Fig. 12. Waveform of the ac/dc and dc/dc stage in the 11-kW OBC and V2L modes. (a) OBC: ac/dc. (b) OBC: dc/dc. (c) V2L: dc/ac. (d) V2L: dc/dc.

emulator and a 3P resistor bank are used in the OBC and V2L modes, respectively. As the battery emulator is used as the load in the OBC mode, the dynamic test was also conducted to validate the stability of the system under varying loads. The waveforms of the dc link voltage and the output current are shown in Fig. 13, from which we can find that the output current was regulated from 0 to 30 A and back to 0 A with a 5-A step. The voltage of the dc link is stable during the whole process, and the system can respond to the charging current

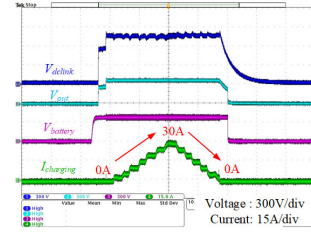


Fig. 13. Dynamic test of the OBC mode with varying output power.

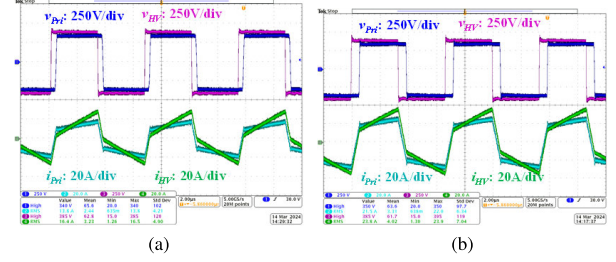


Fig. 14. Waveform of the dc/dc stage in the V2L mode: (a) $R_{\text{phase}} = 34$, $P_{\text{out}} = 4.8$ kW. (b) $R_{\text{phase}} = 22.6$, $P_{\text{out}} = 7.2$ kW.

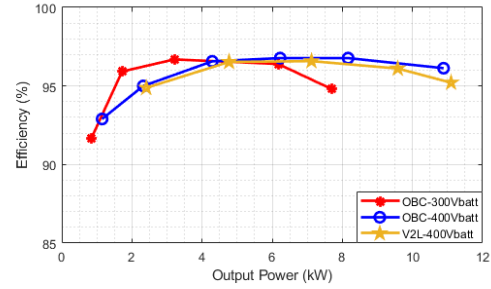


Fig. 15. Efficiency of the OBC and V2L modes with full-power range.

command quickly. In the V2L mode, as the resistor load is used, the tests with varying loads are conducted by changing the resistance of the load tank. The experiment results of two test points with different powers are shown in Fig. 14.

The system efficiency of the OBC and V2L mode with different power is shown in Fig. 15, which includes the efficiency of the OBC mode with 230-V 3P ac input and 300- and 400-V battery voltages, and the efficiency of the V2L mode with 400-V battery voltage and 230-V 3P ac output voltage. The dc current was limited to 25 A. The peak efficiency of 96.9% and 96.6% is achieved in the OBC and V2L modes, respectively, and more than 95% efficiency is achieved at 11 kW. Given the symmetrical and bidirectional features, OBC and V2L modes have very similar efficiencies. These tests validate the performance of the system in OBC and V2L modes.

B. Proposed HM

From Fig. 12, ZVS of all switches can be easily achieved in both OBC and V2L modes by only using the SPS modulation when the load is heavy; however, some of the switches will lose ZVS when the load decreases, which leads to lower efficiency. By using the proposed HM, full-range ZVS and QZVS for all switches of the SHB-FB converter can be

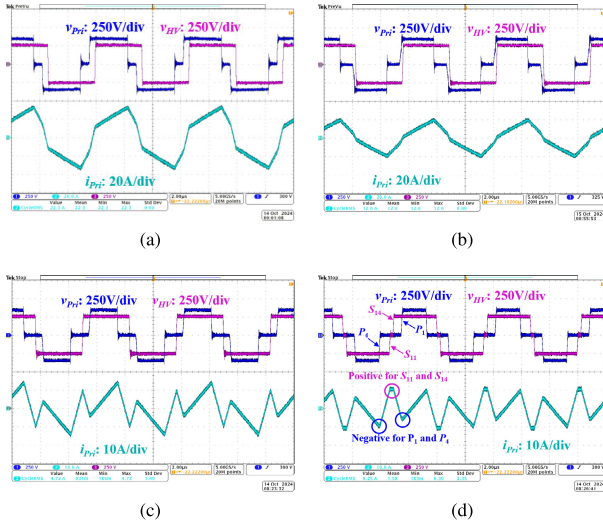


Fig. 16. Transformer voltage and current with 700 V dc bus and 250-V battery using the proposed HM method. (a) State 3: $P_{out} = 5$ kW using PDPS. (b) State 2: $P_{out} = 3.4$ kW using PDPS. (c) State 1: $P_{out} = 1$ kW using PDPS. (d) State 0: $P_{out} = 0.3$ kW using TPS.

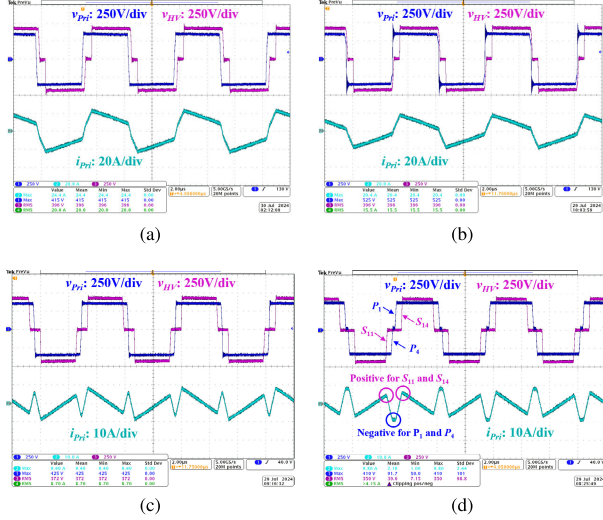


Fig. 17. Transformer voltage and current with 700-V dc bus and 420-V battery using the proposed HM method. (a) State 5: $P_{out} = 5.6$ kW using SDPS. (b) State 6: $P_{out} = 3.8$ kW using SDPS. (c) State 7: $P_{out} = 0.6$ kW using SDPS. (d) State 8: $P_{out} = 0.2$ kW using TPS.

achieved. The effectiveness of the proposed HM was validated by cases of a 700-V dc bus voltage with a 250 and a 420-V HV battery, which are shown in Figs. 16 and 17. Combined with Fig. 12(b)–(d), which shows the typical waveform of state 4, no matter for $V_P > V'_H$ or $V_P < V'_H$, the test results demonstrate that hard-switching is avoided from middle to very light load. And the typical waveform of each state is well aligned with the analysis in Sections III-B and III-C.

The case of charging a 420-V HV battery is explained in detail as an example. When power decreases from full power, the switching current of the primary-side switches will reach the critical value I_r . Then, the state transfers from state 4 to state 5, SDPS takes over, as shown in Fig. 17(a), ZVS is realized for all switches. With power decreases, the switching

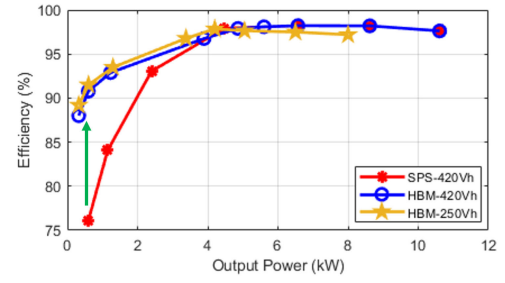


Fig. 18. Efficiency comparison of the SHB-FB converter using SPS modulation and the proposed HM.

TABLE IV
TRANSFORMER VOLUME REDUCTION WITH
THE MAGNETIC INTEGRATION

	This paper	OBC [23]	APM [24]	Reduction
Core	E65/32/27	PQ65/58	PQ50/35	—
Size(mm) $L \times W \times H$	65 × 64 × 54	66 × 52 × 63	50 × 32 × 35	-
Volumn(L)	0.225	0.429	0.056	54%

current of S_{11} and S_{12} also reaches I_r , the state 6 starts, where ZVS for S_{13} and S_{14} and QZVS for other switches can be achieved, as shown in Fig. 17(b). This is because the magnitude of their switching currents is lower than I_r . With a further decrease in power, the switching currents of all other switches except S_{13} and S_{14} reach zero and state seven starts, where the shape of the current waveform changes; however, the ZVS requirement of the current polarity for all switches is still satisfied, as shown in Fig. 17(c). At this stage, the magnitude of the switching current of all other switches except S_{13} and S_{14} increases as power decreases. When they reach I_r again, state 8 starts and TPS takes over. In state 8, the switching currents of S_{13} and S_{14} decrease, and others are fixed to I_r until the power reduces to zero. ZVS for all switches is achieved again in state 8. The case of charging 250 V HV battery is similar. In summary, the proposed HM completely avoids hard switching throughout the full power range and the full battery voltage range.

The efficiency improvement using the proposed HM is addressed by comparing the efficiency of the SHB-FB converter using the conventional SPS modulation and the proposed HM. The experiment results in the power of the SHB-FB converter changing from 0 to 11 kW are shown in Fig. 18. This efficiency improvement will directly result in low loss and, therefore low junction temperature of the GaN devices, which are more subject to thermal runaway compared to SiC/Si devices. Therefore, such efficiency improvement will directly enhance hardware reliability.

C. APM Mode

By adjusting the duty cycle of both the HV and LV sides, the ZVS of all switches can be achieved in the full load range. The transformer waveform of the nominal case, which has the 340-V HV and 14-V LV batteries with the output currents of

TABLE V
COMPARISON OF STATE-OF-THE-ART INTEGRATED OBC AND APM SYSTEMS FOR EVs

	This paper	Standalone OBC [25] & APM [26]	3L DAB [27]	[28]	[3]	[6]	[4]
Integrated components	Tx and switches	No integration	No integration	Tx and switches	Switches	Tx and switches	Tx and switches
Control algorithm	Duty cycle + Hybrid PS	Resonant	TPS	Frequency + SPS	Resonant+PS	TPS	DPS
Topology of three ports	SHB-VFFB-CFFB	OBC: LLC APM: LLC	3L ANPC-ANPC	VFFB for three ports	VFHB-three-leg bridge-VFHB	VFFB for three ports	VFFB for three ports
Bi-directional operation	Yes	Not discussed	Yes	Yes	No	Not discussed	Not discussed
Voltage ranges of three ports (V)	700 / 250-420 / 6-16	400-700 / 240-420 / 6-16	600-800 / 600-800	400 / 270-450 / 9-15	380-700 / 240-330 / 13	400 / 350 / 12	390 / 330 / 12
Transformer turn ratio	6:6:1	OBC: 10:6 APM: 24:1	1:1	29:25:1	OBC: 1.2:1 APM: 6.5:1	20:18:1	26:22:1
HV Switches technology	650V GaN	900V SiC & 650V SiC	650V GaN	650V SiC	900V SiC & 650V SiC	1200V SiC	650V Si
HV Switches amount	8	OBC: 4+4 APM: 4	12	8	2+6	8	8
LV Switches technology	100V GaN	40V Si	-	30V Si	100V Si	80V Si	60V Si
LV Switches amount	4	4	-	4	2	4	4
Peak power of the OBC mode (kW)	11	6.6	3.3	6.6	3.3	3.3	3.3
Peak eff of the OBC mode (%)	97.8	98.5	98.25	97.5	97.8	91.8	95.8
Peak power of the APM mode (kW)	2.6	2	-	2.5	1	1.6	2
Peak eff of the APM mode (%)	95.8	92	-	96.5	95.5	91.8	90
Full-range ZVS of HV DC/DC stage	Yes	No	No	No	Yes	No	No

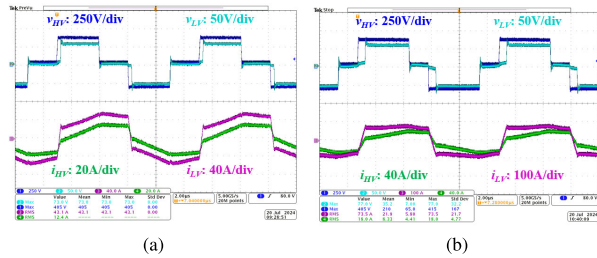


Fig. 19. Transformer voltage and current in the APM mode with a 340-V HV battery and a 14-V LV battery. (a) $I_{out} = 80$ A. (b) $I_{out} = 160$ A.

80 A and 160 A, respectively, are shown in Fig. 19. It should be noted that the current that passes through the switches is a combination of the transformer current and the inductor current, which is the main difference between the CF-DAB and the VF-DAB. Therefore, ZVS can still be achieved even if the transformer current shows the wrong polarity, as shown in Fig. 19(b). The efficiency curves of the APM mode with various voltages of the HV and LV batteries are shown in Fig. 20. These curves indicate that an efficiency exceeding 90% can be achieved, demonstrating the good performance of

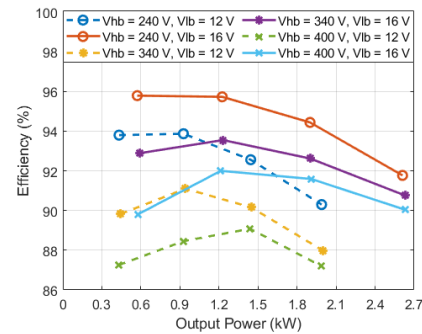


Fig. 20. Efficiency of the APM mode with various input and output voltages.

the APM in a wide voltage range. The peak power of 2.6 kW and the peak efficiency of 95.8% are achieved.

V. CONCLUSION

In this article, a highly integrated charger is proposed and tested with the 11-kW OBC mode, the 11-kW V2L mode, and the 2.6-kW APM mode. The TAB topology achieves the circuit and magnetic integration with a three-winding transformer. Compared to the conventional standalone OBC and APM systems, which need four active bridges and two

pcs of two-winding transformers for the isolated dc/dc stage, the proposed solution only needs three active bridges and one three-winding transformer. At least one active bridge with bulky HV output capacitors and one group of magnetic cores can be saved. A comparison between the proposed integrated three-winding transformer and two transformers used in the standalone OBC and APM, which have similar peak power, is shown in Table IV, which indicates a 54% volume reduction with the magnetic integration. It is worth noting that only one transformer is considered in the APM proposed in [] for fair comparison. What is more, using the SHB converter on the *Pri* to accommodate the 700-V dc bus by 4 pcs of 650-V GaN HMETs not only reduces the number of needed switches compared to other 3L converters but also enables the same level of control freedom compared to the full-bridge converter. Such features are necessary for applying the HM on the SHB-FB converter. Considering the potential of lower GaN device costs in the future and the reduced number of components used, the proposed work shows promise for significant cost reduction.

In addition, an HM method is proposed and applied to the SHB-FB converter in the OBC and V2L modes to improve the efficiency of medium and light power. Using the proposed HM, full-power-range and full-voltage-range ZVS or QZVS can be achieved. Compared with SPS modulation, the light-load efficiency has been improved by more than 10% based on the results of experiments. The comparison between the proposed work and several previous designs for EV charging applications is summarized in Table V. From the comparison, the proposed work demonstrated superiority in both hardware and software design, particularly on magnetic and circuit integration while achieving a full-range ZVS, making it a good candidate for the next-generation EV charging solution.

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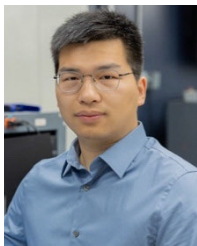


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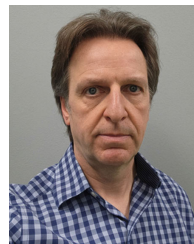


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